Abdullah Sheikh

sheikh07abdullah@gmail.com | (403) 870-6186 | Calgary, AB | LinkedIn | Portfolio

Education

University of Calgary — Calgary, AB
Bachelor of Computer Science — *Expected June 2028*

Technical Skills

Languages: C, C++, Python, Java, Assembly (ARM), Ruby, SQL
Parallel & GPU Computing: CUDA, Metal, OpenMP, ROCm, GPU/SoC Micro-architecture, Memory Hierarchy
Systems & Infrastructure: QNX RTOS, Linux, Embedded Drivers, Interrupt-driven I/O, AWS (PrivateLink, VPC)
Tools & Hardware: Git, GDB, Valgrind, Profilers (Omniperf, ROCprof), Logic Analyzers, Hardware Counters

Experience

Dominion Dynamics — Jan. 2027 – Apr. 2027
Software Engineering Intern — *Ottawa, ON*
- Joining in Winter 2027 to build autonomy systems for Arctic defence.

AMD — Apr. 2026 – Present
Software Engineering Intern — *Calgary, AB*
- Wrote C++ HIP compute kernels for the ROCm stack on Instinct data centre GPUs for HPC/ML applications.
- Improved rocFFT kernel execution times by 5x by using fused Bluestein kernels for prime length FFTs.
- Tripled rocFFT plan capacity by caching JIT-compiled modules, cutting runtime memory ˜40% across pipelines.
- Added SplitMix64 verification to validate bitwise consistency for GPU FFTs, outperforming SHA-256 by 186x.

Shopify — Jan. 2026 – Apr. 2026
Software Engineering Intern — *Toronto, ON*
- Cut circuit breaker CPU ˜20% by profiling and rewriting a hot path in a performance-critical C extension.
- Shipped a fault-injection integration letting 100+ teams validate system behaviour under simulated failures.
- Root-caused incidents across 5+ services, documenting findings in a monitoring tool adopted org-wide.

BlackBerry — Sep. 2025 – Dec. 2025
Software Engineering Intern (QNX) — *Ottawa, ON*
- Shipped 5+ C/C++ drivers on the QNX RTOS, verifying behaviour against hardware functional specifications.
- Designed interrupt-driven I/O on ARM SoCs, hitting sub-millisecond latency across 8+ hardware peripherals.
- Delivered 3x throughput on a C++ pipeline by fixing cache-unfriendly memory access patterns found in GDB.

Enverus — May 2025 – Aug. 2025
Software Engineer Intern — *Calgary, AB*
- Cut $200K+ in annual costs by designing AWS PrivateLink network infrastructure across 350+ VPCs.
- Built 15+ isolated AWS sandboxes, documenting workflows so teams could safely validate infra changes.

Projects

Autopilot Perception Stack | *C++, Python, CUDA, OpenCV* — Jan. 2026
- Built a CNN from scratch in C++/CUDA for real-time lane and obstacle detection on dashcam video streams.
- Optimized CUDA kernels for convolution and pooling layers, hitting 30+ FPS inference on 1080p frames.
- Trained and validated the model on 10K+ labeled road-scene frames, reaching 92% detection accuracy.

Leadership Experience

BobaTalks Mentor — Feb. 2026 – Present
BobaTalks — *Toronto, ON*
- Coaching early-career students through 1:1 sessions on technical growth and industry navigation.

Certifications & Activities

Hackathons: Hack the North @ UWaterloo (2025), Hack the Change @ UCalgary (2025)
Open Source Contributions: QNX Everywhere, Semian, ROCm
AWS: Cloud Practitioner Essentials, Bedrock Workshop, Builder's Quest Badge